ChatGPT gpt-5-3

👤 DEV 3 — ADDITIONS (Route Selection + Green Corridor Orchestration | SynapseSignal)

🔷 🔹 PART 1 — PHASE PLAN (ADDITIONS ONLY)

🔹 PHASE A1 — ROUTE SELECTION ENGINE

Goal: Select optimal route for emergency vehicle

Tasks:

- Receive from Dev 2:
  - multiple routes
  - route metrics (time, congestion)
- Select:

```text
best_route = argmin(estimated_time)
```

- Consider:
  - congestion
  - reliability

Deliverable:
✔ Optimal Route Selection

🔹 PHASE A2 — CORRIDOR GENERATION

Goal: Convert route into signal control plan

Tasks:

- Extract:
  - sequence of intersections

- Map:
  - route → corridor
- Identify:
  - affected intersections

Deliverable:
✔ Corridor Plan

🔹 PHASE A3 — PREDICTIVE SIGNAL SCHEDULING

Goal: Prepare intersections ahead of arrival

Tasks:
For each intersection in route:

- Use:
  - emergency ETA
- Schedule:
  - green phase before arrival
- Ensure:
  - smooth progression

Deliverable:
✔ Predictive Signal Schedule

🔹 PHASE A4 — DYNAMIC CORRIDOR UPDATE

Goal: Adapt to real-time changes

Tasks:

- Monitor:
  - emergency position
  - traffic changes
- Recompute:
  - route (if needed)
  - signal timing

Deliverable:
✔ Adaptive Corridor System

🔹 PHASE A5 — ROUTE VALIDATION (SUITABILITY CHECK)

Goal: Ensure selected route remains optimal

Tasks:

- Continuously evaluate:
  - congestion changes
- Compare:
  - current route vs alternatives
- Trigger:
  - reroute if needed

Deliverable:
✔ Route Validation Engine

🔹 PHASE A6 — GREEN CORRIDOR EXECUTION

Goal: Apply real signal overrides

Tasks:

- Activate:
  - green signals along route
- Maintain:
  - safe transitions
- Clear:
  - conflicting flows

Deliverable:
✔ Corridor Execution Logic

🔹 PHASE A7 — POST-PASSAGE RESTORATION

Goal: Return system to normal

Tasks:

- Detect:
  - emergency vehicle passed
- Restore:

    - normal signal logic
- Rebalance traffic

Deliverable:
✔ Restoration System

🔹 PHASE A8 — CORRIDOR OUTPUT API

Goal: Provide data to Dev 4 & Dev 5

Output format:

```json
{
  "route": [...],
  "active_corridor": [...],
  "current_intersection": "...",
  "eta_sequence": [...],
  "status": "active/replanned"
}
```

Deliverable:
✔ Corridor API

🔷 🔹 PART 2 — MASTER PROMPTS (ADDITIONS ONLY)

1️⃣ MASTER PROMPT — ROUTE SELECTION ENGINE

You are designing routing logic.

Task:
Select optimal route for emergency vehicle.

Input:

- routes with metrics

You must:

- Select route with:
    - minimum estimated_time
- Consider congestion

Output format:

```json
{
  "selected_route": "...",
  "reason": "lowest_time"
}
```

Constraints:

- Must be deterministic
- No arbitrary choice

2️⃣ MASTER PROMPT — CORRIDOR GENERATION

You are building corridor logic.

Task:
Convert route into sequence of intersections.

You must:

- Extract nodes
- Define corridor path

Output format:

```json
{
  "corridor": ["I1", "I2", "I3"]
}
```

Constraints:

- Must follow valid graph path

3️⃣ MASTER PROMPT — PREDICTIVE SIGNAL SCHEDULING

You are designing timing logic.

Task:
Schedule signals based on ETA.

You must:

- For each intersection:

    - compute arrival time
    - schedule green

Output format:

```
json

{
  "intersection_id": "...",
  "green_start": ...,
  "green_duration": ...
}
```

Constraints:

- Must align with movement timing
- No unsafe transitions

4️⃣ MASTER PROMPT — DYNAMIC CORRIDOR UPDATE

You are designing adaptive routing.

Task:
Update corridor in real-time.

You must:

- Monitor:
    - position
    - congestion
- Trigger reroute if needed

Output format:

```
json

{
  "updated_route": [...],
  "reason": "congestion_change"
}
```

Constraints:

- Must avoid frequent oscillations
- Must be stable

5️⃣ MASTER PROMPT — ROUTE VALIDATION

You are validating route quality.

Task:
Check if current route is still optimal.

You must:

- Compare:
  - current vs alternatives
- Trigger reroute condition

Output format:

```json
{
  "is_optimal": true/false,
  "better_route": "..."
}
```

Constraints:

- Must be continuous evaluation
- No false switching

6️⃣ MASTER PROMPT — CORRIDOR EXECUTION

You are controlling signals.

Task:
Activate green corridor.

You must:

- Turn signals green in sequence
- Prevent conflicts

Output format:

```json
{
  "intersection_id": "...",
  "override": true,
  "state": "GREEN"
```

```
}
```

Constraints:

- Must be safe
- Must follow schedule

7️⃣ MASTER PROMPT — RESTORATION SYSTEM

You are restoring normal control.

Task:
Return system to normal after passage.

You must:

- Detect passage completion
- Restore signal logic

Output format:

```json
{
  "intersection_id": "...",
  "mode": "normal"
}
```

Constraints:

- Must be smooth transition
- No abrupt changes

8️⃣ MASTER PROMPT — CORRIDOR API

You are exposing system state.

Task:
Provide corridor data to frontend.

Output format:

```json
{
  "route": [...],
```

```
  "active_corridor": [...],
  "status": "active"
}
```

Constraints:

- Must be real-time
- Must be consistent

🔷 🔹 PART 3 — GUIDED LEARNING (ADDITIONS ONLY)

🔹 PHASE A1 — ROUTE SELECTION

🧠 What You Learn

You learn **how systems choose optimal paths under constraints**.

⚙️ What Happens

- Evaluate multiple routes
- Select best one

🔍 Key Insight

Shortest ≠ fastest

🧱 Mental Model

Routing is optimization, not distance minimization

🔹 PHASE A2 — CORRIDOR GENERATION

🧠 What You Learn

You learn **how paths become control plans**.

⚙️ What Happens

- Convert route → intersections

🔍 Key Insight

Control happens at intersections, not roads

🧱 Mental Model

Route → control points

🔹 PHASE A3 — PREDICTIVE SCHEDULING

🧠 What You Learn

You learn **anticipatory control systems**.

⚙️ What Happens

- Prepare signals before arrival

🔍 Key Insight

Reactive control is too late

🧱 Mental Model

Control must predict, not react

🔹 PHASE A4 — DYNAMIC UPDATE

🧠 What You Learn

You learn **adaptive systems under uncertainty**.

⚙️ What Happens

- Adjust route dynamically

🔍 Key Insight

Traffic is non-static

🧱 Mental Model

System must adapt continuously

🔹 PHASE A5 — ROUTE VALIDATION

🧠 What You Learn

You learn **continuous optimization**.

⚙️ What Happens

- Validate route quality

🔍 Key Insight

Best decision now may not remain best

🧱 Mental Model

Optimization is ongoing

🔹 PHASE A6 — CORRIDOR EXECUTION

🧠 What You Learn

You learn **real-world actuation**.

⚙️ What Happens

- Signals are overridden

🔍 Key Insight

Execution must be safe

🧱 Mental Model

Control = safe actuation

🔹 PHASE A7 — RESTORATION

🧠 What You Learn

You learn **system recovery after override**.

⚙️ What Happens

- Return to normal flow

🔍 Key Insight

Recovery is as important as execution

🧱 Mental Model

Override must be temporary

🔹 PHASE A8 — CORRIDOR API

🧠 What You Learn

You learn **how backend communicates complex state visually**.

⚙️ What Happens

- Send route + corridor

🔍 Key Insight

Visualization depends on clean data

🧱 Mental Model

Backend defines what frontend can show

🧠 FINAL INSIGHT

> These additions transform Dev 3 from a controller
> into a **predictive, route-aware orchestration engine**

✔ Now supports:

- Green corridor system
- Route optimization

- Real-time rerouting
- Map visualization integration